

AI Detectors and Ad Hominem Argumentation

Structural Kinship of Old and New Manipulation Patterns

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Preamble

This text grew out of a discussion on the mailing list of the *Information Physics Institute* (May 2026). Peter Austin reported that his own emails are regularly marked by AI detectors as more than 50% AI-generated – even when written in personal correspondence and without any AI assistance. The initial question in the group was practical: what to do about these false positives.

On closer examination, the question turns out to be not primarily technical but methodological and psychological. AI detectors are not the beginning of a new phenomenon. They stand at the provisional end of an old line: the argumentation patterns that reject without engaging with the substance. This document describes the structural kinship and names the psychological mechanism that drives the pattern.

1 The contradiction in the demand

At the centre of the current debate stands an open contradiction:

- On one side we demand precise, error-free, well-structured texts. This has always been the standard of scientific and professional communication.
- On the other side, *the absence of errors* is now itself suspicious. A text with low perplexity, uniform sentence structure, and clean transitions is flagged by detectors as AI.

Taken literally, this is an **instruction to write worse**: produce enough roughness to pass through, enough variation to avoid being read as a machine. This cannot be a methodological standard.

The evidence for this skew is well documented. The Stanford study by Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu and James Zou – *GPT detectors are biased against non-native English writers* (Patterns, Vol. 4, No. 7, Article 100779, 14 July 2023; arXiv:2304.02819) – showed: seven widely used GPT detectors classified *more than half* of TOEFL essays by non-native English learners as “AI-generated”, while the same detectors achieved near-perfect accuracy on US 8th-grade essays. The reason: non-native texts typically have lower perplexity, because vocabulary is more limited and sentence structures simpler – exactly the feature they share with machine-generated text.

OpenAI itself – the maker of ChatGPT – published in January 2023 an *AI Classifier* that was supposed to solve precisely this problem. Six months later, on 20 July 2023, the tool was withdrawn. The official reason given by OpenAI: only 26 % of genuinely AI-generated texts were correctly identified, and in 9 % of cases human texts were falsely flagged as AI. With a hit rate below 50 %, a detector is worse than a coin flip.

Anyone who writes carefully as a technician, scientist, or non-native speaker necessarily produces clearer and

more structured text than a casual writer. It is precisely this writing that is systematically flagged.

2 Substance, not style

The real distinction between human and machine-generated content does not lie in style, but in substance.

AI texts are characterised not by their form, but by a specific substantive weakness: they are often coherent-sounding arguments built on *hallucinated facts*, *fabricated citations*, and *confidently presented but vacuous derivations*. The linguistic polish is real; the substance is invented.

A critical reader who understands the subject recognises this weakness – not from surface markers, but because the claims fail on closer inspection. A statistical machine cannot in principle see this difference, because it is not a surface feature.

Consequence: whoever wants to judge content must be able to understand content. Detectors cannot do this. They can only count symptoms.

3 The detector user as the actual misuser

Here an inversion appears that is rarely articulated in the debate. Whoever uses an AI detector instead of reading the text themselves **delegates their own judgement to an algorithm**.

The detector user is thereby more deeply dependent on AI than the author who consciously uses AI as a tool:

Role	What the person does	Who holds the judgement
Author with AI tool	substance their own, language with AI help	human
Reviewer with detector	reads score, takes judgement from machine	machine

The reproach “you used AI” therefore often comes from exactly the person who has already handed over their own critical function to AI. They accuse the author of what they themselves do – without being aware that they are doing it.

4 AI as filter, not as oracle

An important distinction is needed. Even careful knowledge workers use AI – for summaries, for language formalisation, for pre-sorting large amounts of text. This is not the problem. The problem is *the kind* of use.

AI as filter (triage):

- AI reads large amounts of text and summarises.
- The human reads the summary and decides: does this match, or do I need to look more closely?
- In case of ambiguity or unusual findings: the human goes into the detail.
- The final judgement always rests with the human.

AI as oracle (delegation):

- AI delivers a score, a classification, a truth.
- The score is accepted.
- No own examination of substance.
- No willingness to read for oneself in case of irregularities.

AI use as **filter** is methodologically unproblematic – it corresponds to the long-established practice of having larger amounts of text pre-sorted by abstracts, bibliographies, or assistants. AI use as **oracle** is methodologically fatal: it replaces judgement instead of supporting it.

Detectors are the paradigm case of oracle use. They deliver a number that can be accepted or ignored – but no substance that could be examined.

5 Structural impossibility of substantive judgement

Here lies the technical point that pulls the rug from under the entire detector construction.

An AI detector implicitly claims: “I can recognise whether a text comes from an AI.” For this it would have to do something that AI itself cannot do:

- AI does not know whether a claim is true.
- AI does not know whether an argument holds.
- AI does not recognise its own hallucinations.

If AI itself is overwhelmed at the substance level, a detector – which is also AI – cannot judge substance either. It can only measure surface features: word probabilities (perplexity), sentence-length variation (burstiness), frequency of certain transition words.

The construction dissolves itself logically:

Whoever wants to judge substance must be able to understand it. Detectors are AI. AI cannot understand substance. So detectors cannot judge what they claim to judge.

A detector that could really judge substance would no longer be an AI – it would be a *critically thinking human with subject expertise*. But that is precisely what

the detector is supposed to replace. The claim is self-contradictory.

5.1 An analogy

No one would believe someone who said: “I can tell whether a maths book is written by an incompetent author – without knowing any mathematics myself. I just measure the frequency of certain words.” But this is exactly what AI detectors do: they claim to judge **substance** through **form statistics**.

This is the same category error as in phrenology (skull shape equals character) or polygraphs (sweat production equals truth): *measuring surface correlates and presenting it as measurement of the underlying quantity*.

6 Kinship with ad hominem argumentation

Up to here the argument has been technical. Now comes the structural point: AI detectors are not a new phenomenon.

They do to texts what ad hominem argumentation does to people.

Ad hominem argumentation	AI detector
attacks the speaker, not the argument	rates the form, not the content
who said it?	who (or what) wrote it?
bypasses substantive engagement	bypasses substantive examination
convenient – no understanding needed	convenient – no reading needed
replaces thinking with suspicion	replaces reading with a score

Both follow the same pattern: **shifting evaluation from the matter itself to a property of the source**. Instead of testing whether something is right, the question becomes where it comes from – and from origin a conclusion about validity is drawn.

In logic this has been known for centuries as the *genetic fallacy*. Aristotle described ad hominem arguments in his *Sophistical Refutations* as fallacies. Arthur Schopenhauer addresses the related *argumentum ad personam* in his *Eristic Dialectic* as a pseudo-argument that bypasses the matter. The mechanism is 2400 years old; the technical completion in software form is new.

7 Cognitive dissonance as the driver of the pattern

Why does this pattern work so reliably? The answer comes from Leon Festinger's *A Theory of Cognitive Dissonance* (Stanford University Press, 1957), one of the most influential works of 20th-century social psychology.

7.1 Festinger's basic observation

Festinger developed the theory from an unusual empirical observation. In *When Prophecy Fails* (with Henry W. Riecken and Stanley Schachter, University of Minnesota Press, 1956) he documents the behaviour of a doomsday cult that had predicted a concrete date for the end of the world. When the date passed without anything happening, the members' belief *strengthened* rather than collapsing. They interpreted the failure of the prophecy as proof that their faith community had saved the world.

Festinger's conclusion: when two convictions are incompatible, psychological pressure (*dissonance*) arises. This pressure does not necessarily lead to correction of the false belief – often it leads to *reinterpretation of reality* so that the original position remains confirmed.

7.2 Application to the detector context

In the AI-detector situation, the following dissonance typically arises in the reviewer:

Conviction A	Conviction B (triggering)
“I am a competent reviewer”	“Before me lies a text clearer than what I would write”
“I belong to those who think critically”	“Someone here is apparently thinking more clearly than I am”
“AI makes people dumber”	“But this text does not seem dumb – it seems cleverer than usual”

The dissonance is uncomfortable. There are two ways to resolve it:

Path 1: Substantive engagement (strenuous).
Reconsider one’s own position; acknowledge that the other has done something good; adjust one’s own standards; revise one’s own position if necessary.

Path 2: Source devaluation (convenient).
Mark the text as inauthentic – “this is not really from them, this is AI”. Self-perception remains intact. No own revision needed.

The AI detector provides the *technical justification* for Path 2. It allows the uncomfortable text to be disqualified without having to engage with it. The score looks scientific, objective, neutral – and provides exactly what the reviewer needs to resolve the dissonance without questioning themselves.

7.3 Three functions in one

Dissonance reduction through source devaluation fulfils three functions simultaneously:

- **Self-esteem protection:** “I am not less competent – they just cheated.”
- **Effort avoidance:** “I do not have to engage with this.”
- **Social positioning:** “I am one of the critical ones who spot AI cheating.”

All three functions are *cheap* to obtain – a single click on a detector. The honest alternative – substantive examination – is *expensive*: time, concentration, the risk of having to revise one’s own position.

7.4 Amplification through pseudo-objectivity

Classical ad hominem arguments can be directly rebutted by the addressee – one can reply, contextualise, defend oneself. AI detectors have three additional features that make dissonance reduction even more effective:

1. **Pseudo-objectivity:** The number looks scientific, not personal. The reviewer can say: “I am not attacking you – the algorithm says this.”
2. **Responsibility offloading:** Nobody feels personally responsible for the verdict. “The computer decided.”
3. **Irrefutability:** The addressee *cannot prove* that they did not use AI. Negation is not provable.

These three features make the AI detector an *ideal tool for dissonance reduction*: maximum effect with minimal personal investment and minimal responsibility.

7.5 A bitter point

From this follows a paradoxical consequence: the **clearer and more competent** a text is, the **greater the dissonance** for a less self-assured reader – and the **stronger the motive** to mark the text as AI. Detectors thus punish exactly the competence they claim to protect. Not by accident, but by psychological function.

8 The whole family of rhetorical manipulation patterns

The kinship is not confined to the ad hominem fallacy. AI detectors in fact reproduce a *whole family* of classical manipulation techniques – each in technical form.

8.1 Tone policing

Human: “You are getting emotional – one cannot take this seriously.”

Detector: “You write too smoothly – one cannot take this seriously.”

Style becomes the disqualifier, regardless of content.

8.2 Strawman argument

The *human* distorts the opponent’s position into a simpler, easier-to-refute caricature. The *detector* reduces the text to statistical surface features that have nothing to do with the actual content. In both cases a simplified caricature is judged, not the original.

8.3 Burden-of-proof reversal

Human: “Prove to me that you thought this up yourself.”

Detector: “Prove to me that you didn’t use AI.”

In both cases the opposite is taken as default. The addressee has to prove their innocence – which is structurally impossible because *non-existence is not provable*.

8.4 Argumentum ad verecundiam

Human: “Professor X said so, therefore it must be true.”

Detector: “The algorithm says 87 %, therefore it must be true.”

An authority is placed above the content. With AI detectors the authority is a machine – which sharpens the inversion.

8.5 Argumentum ad populum

Human: “Everyone knows that...”

Detector: “This looks like typical AI style, all the models recognise the pattern.”

Consensus (or learned pattern) replaces argument.

8.6 Gaslighting

Human: “You overestimate yourself – this cannot come from you.”

Detector: “The score shows: this cannot come from you.”

The addressee is denied their own authorship. They are made to doubt themselves. This is the most destructive form – it targets not the claim, but the person.

9 The historical line

The patterns described here are not new. They are documented in every epoch:

- **Ancient rhetoric:** Aristotle described ad hominem arguments in the *Sophistical Refutations* as fallacies.
- **Middle Ages:** *Argumentum ad auctoritatem* (appeal to authorities instead of own examination) as a scholastic argumentation form.
- **Inquisition:** suspicion logic – whoever thinks like this is suspect.
- **20th-century totalitarianism:** “whoever speaks like this is an enemy.”
- **Present:** AI detectors – “whoever writes like this is not real.”

It is always the same pattern: when substantive engagement becomes difficult, one falls back on source criticism. The **historical constancy** is the most important observation of this document – it shows that the real problem is not technological but *anthropological*.

10 What would actually help

If detectors do not help but worsen the problem, what does?

Substantive reading. A competent reader who understands the subject recognises hallucinations because they understand the substance. No score required. What is required is willingness to engage with the content.

Reproducible verification. Where content is checkable, it can be checked – through code, data, scripts that one can run oneself. This is not a detector function; this is a *content function*.

Provenance documentation. Drafts, voice recordings, revision history. Anyone who makes their working process transparent shifts the question from “prove you didn’t use AI” to “here is the workflow, see for yourself”. The second question can be answered.

Open discipline. Use AI consciously as a tool, not as an oracle. Mark your own position as your own. Acknowledge AI assistance openly rather than hiding it.

None of these points needs a detector. All four need *competent, honest people who are willing to do the work of understanding*.

11 Diagnosis

The AI-detector debate is not primarily technical. It is the visible peak of a cultural tendency that replaces the strenuous business of understanding with the convenient practice of source criticism.

In one sentence:

Ad hominem argumentation is the AI detector for people. The AI detector is ad hominem argumentation for texts. Both are the refusal of the matter, dressed up as concern for the source.

The psychological driver – cognitive dissonance and its resolution through source devaluation – has been well documented since Festinger 1956/1957. The historical constancy of the pattern reaches back to Aristotle. The technical completion in software form is new; the underlying problem is old.

Whoever trusts an AI checker more than their own critical judgement has ceased to be a reviewer – and has themselves become an appendage of a machine. That is the real misuser of AI.

12 Sources

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Note on the genesis of this document. The text was written in May 2026, prompted by a discussion on

the mailing list of the *Information Physics Institute* with Peter Austin. The argument is my own thinking. Language formalisation was done with AI assistance – in exactly the manner described in §5 as *AI as filter, not as oracle*. An open discipline: substance from the author, language with tool support, judgement with the human.